Thesis Management Plan & Personal Reflection

Part I: Thesis Management Plan

1. Research Objectives, Goals, and Scope

Title & Core Research Focus

- *Thesis Title:* Developing industry-ready automated trading systems to conduct EcoFin analysis using deep learning algorithms (IRP 12).
- *Core Research Agenda:* Development and optimization of multi-agent automated systems for financial trading, focusing on the seamless integration of time series, statistical models, fundamental variables, and Large Language Model (LLM)-based financial document analysis.
- *Advanced Decision-Making Architectures:* Formulation, training, and benchmarking of reinforcement learning (RL) agents and deep learning frameworks for automated execution and portfolio optimization.

Primary Research Objectives & Associated Studies

- *High-Frequency Market Microstructure:* Analysing how external information shocks impact market microstructure. Conducted nanosecond-scale market microstructure analysis following U.S. macroeconomic news releases and central bank announcements at major exchanges (Eurex and CME). Assessing the impact of ultra-low latency execution on price discovery, volatility, trading volume, and liquidity provision.
- *Long horizon price impact modeling:* Modeling long horizon price impact, trade-level Profit & Loss (P&L) across liquidity tiers, benchmarking institutional execution strategies, and computing excess alpha by incorporating proprietary structural variables such as portfolio order types and full orderbook state.
- *Multimodal Agentic Frameworks & Risk Forecasting:* Developing multimodal vision-language architectures for financial anomaly detection and investigating zero-shot Value-at-Risk (VaR) and Expected Shortfall (ES) forecasting with conformal calibration in Central and Eastern European (CEE) markets.

Key Milestones & Deliverables

- *Milestone 1 (Foundational Infrastructure & Early Publications):* Construction of high-frequency data pipelines, foundational coursework in machine learning and optimization, and initial conference publications (e.g., *LLM-Driven Stock Prediction: Capturing Market Trends with LLaMA*, ICBE 2025).
- *Milestone 2 (Industrial Secondment & Advanced Modeling):* Execution of industrial secondment at Deutsche Börse AG and Eurex focusing on order flow valuation, liquidity analytics and alpha generation. Submission of core journal articles and conference articles. Finish the secondments and modeling by mid of 2027.
- *Milestone 3 (Synthesis & Defense):* Integration of empirical findings, completion of working papers into finalized journal submissions, and compilation of the doctoral dissertation by end of 2027.

2. Data Management Plan (DMP)

Data Acquisition and Heterogeneous Sources

- *High-Precision Exchange Datasets:* Collection and processing of high-precision timestamped order flow data, limit order book records, and execution logs from primary institutional venues including Eurex and CME.
- *Equities and Market Indices:* Acquisition of comprehensive datasets from Refinitiv, Yahoo Finance, and DAX and MDAX components traded via Xetra.
- *Alternative and Textual Data:* Integration of central bank communication archives, macroeconomic news feeds, and corporate financial disclosures to support LLM-driven sentiment analysis and policy impact evaluation.

Storage Architecture, Security, and Version Control

- *Codebase and Quantlet Management:* Utilization of GitHub repositories and Quantlet platforms for version-controlled quantitative scripts, algorithmic notebooks, and reproducible research artifacts.
- *Collaborative Manuscript Development:* Secure cloud-based document authoring and collaborative editing via Overleaf (e.g., project workspaces linked to ongoing exchange and market efficiency studies).
- *Proprietary Governance:* Strict compliance with data confidentiality protocols and governance guidelines established during industrial secondments (Deutsche Börse AG) to protect sensitive order flow metrics and proprietary portfolio structures.

Open Science & Reproducibility Standards

- *Quantlet Repository and Quantinar Course Publication:* Systematic release of computational code, processed data, expalanatory presentations and empirical notebooks to foster academic transparency and replicability across institutional research groups.

3. Resources

Academic and Industrial Supervision Structure

- *Academic Supervisory Committee:* Prof. Daniel Traian Pele, Prof. Wolfgang Karl Härdle, and Prof. Adrian Costea.
- *Industrial Supervisor:* Dr. Stefan Schlamp (Deutsche Börse AG).

Advanced Training Schools and Institutional Networks

- Active participation in specialized European doctoral training networks and schools, including:
    - MSCA Training School on Ethics in AI and Digital Finance at the Bucharest University of Economic Studies (ASE).
    - PhD Training Week on Reinforcement Learning in Finance at the University of Twente.
    - Training modules on Agentic AI application in Finance (Raiffeisen Bank) and ML for tabular data analysis (Cardo AI) and Training school on Data Science in Digital Finance hosted at Babeș-Bolyai University, Cluj.

Computational Infrastructure

- High-performance computing (HPC) nodes equipped with multi-GPU accelerators to handle heavy tensor operations required for transformer-based LLMs, vision-language models, and deep reinforcement learning training environments. API access to use paid state of the art proprietary models.

4. Risk Analysis and Proactive Mitigation Strategies

Anticipated Risk	**Likelihood**	**Impact**	**Proactive Mitigation Strategy**
High-Frequency Data Bottlenecks: Managing massive nanosecond-scale order book logs can saturate local storage, slow the modeling and memory bandwidth.	Moderate	High	Implement modular data chunking, streaming data architectures, and distributed computing on university computational resources.
Model Overfitting in Financial Regimes: Multi-agent reinforcement learning models and LLMs risk memorizing historical market noise rather than robust trading rules.	High	High	Enforce rigorous out-of-sample cross-validation, walk-forward testing across distinct macroeconomic shock periods, and strict economic regularization.
Industrial Secondment Operational Friction: Balancing intense empirical academic research with industrial usecase milestones during the Deutsche Börse AG placement.	Low	Moderate	Utilize agile sprint planning and maintain synchronized alignment meetings between academic supervisors and industrial mentors.

5. Project Timeline & Management Methodologies

- *Management Methodologies:* Deployment of Gantt charts for high level milestone tracking, resource allocation frameworks for compute budgeting, and continuous risk mitigation plans.
- *Timeline Overview:*
    - *2024 end:* Foundational coursework in data mining, optimization, non-parametric statistics, and academic ethics; initial conference presentations.
    - *2025:* Advanced empirical research; participation in specialized MSCA training weeks; industrial secondment at Deutsche Börse AG; Market microstructure and order flow papers; Collaborating with other colleagues on modeling task.
    - *2026:* Finalization and submission of first journal papers and additional conference proceedings; Gather additional data; Modeling and drafting papers for final research publication.
    - *2027:* Finalization all journal papers submission; Drafting the thesis; Look for future positions; Thesis defense.

Part II: Personal Reflection on Project Management Concepts

Critical Analysis of Research Practice and Organizational Growth

Engaging with formal project management methodologies like structured milestone planning, resource scheduling, critical workflow analysis, and risk mitigation matrices has helped fundamentally reshape and structure the architecture and execution of my doctoral research. At the intersection of high-frequency financial engineering, deep reinforcement learning, and natural language processing, I faced and am sometimes still facing an extreme risk of scope creep. The abundance of potential research directions, from nanosecond-scale order book microstructure to macroeconomic LLM analysis, necessitates rigorous operational discipline. Adopting structured project management principles allowed me to transition from an ad-hoc experimental approach to a systematic, milestone-driven workflow. By dividing the doctoral lifecycle into distinct operational phases ranging from foundational data pipeline engineering at Eurex and CME to advanced multimodal model integration has helped me effectively manage complex dependencies. For instance, framing my industrial secondment at Deutsche Börse AG not merely as an external placement, but as a tightly scoped research sprint with explicit deliverables (such as exchange-paid order flow valuation models), ensured that practical industry insights directly reinforced my academic publications.

Furthermore, applying risk analysis transformed how I handled technical roadblocks. In empirical financial data science, hardware latency, data cleaning anomalies, and model convergence failures are inevitable. Constructing a proactive risk matrix enabled me to anticipate bottlenecks before they stalled experimental progress, facilitating the creation of robust fallback strategies and scalable cloud-based computing pipelines.

Finally, this reflective practice underscored that academic research relies as heavily on rigorous project organization as it does on theoretical innovation. The management methodologies internalized throughout this process is supporting me in streamlining the successful delivery of my PhD commitments but also establishing durable, professional frameworks for leading complex, cross-functional initiatives in future academic or industry environments.